

Maintenance sheet version 10 for ablation area (tripod, one level)

Station name	NUK_K
People present	PHO, ACH (Asian)
Purpose	Maintenance / installation / take-down / quick visit / emergency
Date	21/09/23

Notes:

AWS to do list		Check after maintenance			
Part	Shipped (Y/N)	Replaced? (Y/N)	Number shipped/arrival	Number departure	
Radiometer (write down the serial number you use!)	Y	Y	S: 212889/150711	212889	Clean? Y/N
Ablation hose assembly	N	N			No air in hose and bladder half full? Y/N
Iridium modem IMEI number	In loggerbox	Y	S: 300534063817730 1299770	30053	4062923450
Inclinometer/compass	In CNR4	Y	Attached to the radiometer? Y/N		
Satellite antenna	N	N			
Wind sensor	N	N	Black box pointing toward the radiometer (i.e., along the boom)? Y/N		
Temperature / humidity assembly	Y	Y	Fan spins? Y/N Sounds, OK? Y/N		
HygroClip T/hum assembly	N	N	NB: Rotronics (Y/N) could have a 40 °C offset. New or old type? NB: Luft sensor (Y/N)		
Sonic ranger on AWS	Y	N	Old sensor with new membrane? Y/N		
Sonic ranger on stake(s)	Y	Y	Old sensor with new membrane? Y/N NB: make sure not being buried next winter		
Thermistor string	Y (incl coupler)	Y			
Solar panel	N	N	Clean? Y/N NB: is not effective when partially covered		
GPS antenna	In loggerbox	Y	At top of logger enclosure? Y/N		
Logger box ID	T3B-0005		CR1000 or CR1000X? ID number: T3B-0004		
Multiplexer	N	N	Clicking during measurement cycle? Y/N		
Logger enclosure	Y	Y	Replaced (including everything inside)? Y/N New desiccant bags? Y/N		
Battery box	N	N	New box? Y/N New desiccant bags? Y/N		
Stakes	N	Y	New one(s) drilled in? Y/N Length of new stake(s) 8 m? Y/N Bottom plugged to reduce melt-in? Y/N		
Tripod	N		New tripod? Y/N New sensor boom? Y/N		

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Metadata maintenance

	Arrival	Departure
Time difference between logger clock and UTC ($\pm$ mm:ss)	N/A	1.5
Download data and/or change CF/SD Card (wait for green light)	will do back home Y/N	Y/N
Name of logger program	N/A	PROMICE v2.2
Screen dump of values in fast scan mode	Y/N	Y/N
Photos of tripod, sensors, logger box wiring etc. (more is better)	Y/N	Y/N
Mast tilt towards radiometer (- if tilting towards anemometer)	-4°	-2.5°
Mast tilt across boom looking from radiometer (- if clockwise)	-7°	-2.6°
Boom direction relative to north	29° NE true/magn	26° NE true/magn
Box on wind sensor exactly towards mast? If not, measure.	Y/N	Y/N
Radiometer exactly aligned with inclinometer? If not, measure.	Y/N	same unit
Temperature / humidity height (bottom of casing - surface)	225 cm	260 cm
Stake assembly standing?	NO SR50 Y/N / sort of	Y/N / sort of
Free length of stake (if 3: outer, holding boom with sonic ranger)	238 cm	158 cm
Free length of stake 2 (if 3: middle)	surface to top of stake 142 * cm	142 * cm
Free length of stake 3 (if 3: outer, without sonic ranger)	dist to SR50 135 cm	135 cm
Height of ablation bladder fluid level above the ice surface	115 cm	cm
Vertical difference in surface height (at station - at hose) (approx.)	cm	cm
Length of thermistor string on surface from surface marking	dist to h 15 therm. 200 cm	cm
Apparent damage to sensors? If yes, which?	Y/N	Y/N

Notes:

- NUK-V logger box used instead of NUK-K * bottom of cross beam (NUK-K is now placed at NUK UV3)

Must-do at next visit:

- Replace SR50 on stake with a longer cable. The current one is a little short, but we had no other cable.
- Station should be placed on flatter ground and sensors remounted.
- Temp/humidity Inft sensor should be mounted beneath boom. We couldn't tilt the station to do this, so mounted it on the boom this time.

- Inft sensor also had a chip in the lower part of the casing. Will this affect measurements?